

Model-Free Implied Variance Measures

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Model-free implied variance is typically based on the log-contract introduced by Neuberger (1994). From the log-contract there are at least two ways of characterizing the variance of future returns. We analyze the difference between these two approaches in a general semimartingale setup and show that the difference is mainly characterized by the volatility of stochastic variance and the correlation between the processes for stochastic variance and asset prices. Using option prices for the DAX we perform an empirical analysis for the difference both variance measures and show that it can be explained to a large extent by the variance swap rate. Furthermore, we show for the Heston model that one can take advantage of the difference for estimating and calibrating the model.

JEL classification: G12; G13

Keywords: log-contract; variance risk; implied risk-neutral density; Heston model

1 Introduction

The information content of option prices is a subject of current research for some time. It is well-known from Breeden and Litzenberger (1978) that plain vanilla option prices for a continuum of strikes explicitly describe the risk-neutral pricing measure. Since

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in practice, only a finite number of traded options is available, one might ask what information about the risk-neutral distribution is still in existence. For example, implied volatilities from the Black-Scholes model derived from traded options can be used for the calibration of an arbitrary model. Since it is well documented that volatilities are itself stochastic, in a more general setup, the option prices can be used for calculating variation measures under the pricing distribution. In particular, variance indices like CBOE's Volatility Index (VIX) or Deutsche Börse's VDAX are calculated from option contracts. In the beginning these indices used only at-the-money options and simply represented the implied Black-Scholes at-the-money volatility. If the Black-Scholes is correct, this method would be sufficient and gives an efficient forecast of future volatility. But, typically we observe volatility smiles and smirks. Additionally, empirical properties of asset returns makes clear that the assumptions of the Black-Scholes model cannot be correct.¹.

Today's construction of these indices is based on a broader volatility surface, making use of all information from option markets. The variance indices are derived from a static portfolio of out-of-the-money vanilla options. The static portfolio represents a hedge for the so-called log-contract, which measures the return of an asset.² Dupire (1993) and Neuberger (1994) show that the log-contract can be used to derive a model-free implied variance measure. Furthermore, Demeterfi et al. (1999) and Britten-Jones and Neuberger (2000) show that this implied variance measure equals the true risk-neutral expectation of integrated stochastic variance. Although the original derivation assumes that the underlying price path is continuous, similar results can be achieved for more general models. Jiang and Tian (2005) and Carr and Wu (2009) shows that this relation is also approximatively correct for jump diffusion and semimartingale models. Heuristically, all derivations make use of Itô's lemma and consider the quadratic variation of the asset returns. Under the risk-neutral measure the quadratic variation also represents the drift of the log-contract. Therefore, the expectation of log-contracts already presents information about expected realized variance. Jiang and Tian (2005) uses this methodology in order to forecast future realized variances and concludes that the model-free implied variance subsumes all information contained in B-S implied volatility and historical volatility.

A fundamental application for this variance measure is the opportunity to trade realized variation and volatility. For example, variance swaps consist of two payment streams.

¹See, e.g., Cont (2001)

²See Neuberger (1994)

The floating leg is usually computed by summing the squared high-frequency returns. The difference to realized quadratic variation is then given only by the sampling frequency, see Sepp (2008). The expected risk-neutral realized variation represents the fixed leg and is called the variance swap rate. The differences between realized and expected risk-neutral variation then quantify variance risk premiums. Carr and Wu (2009) investigate variance swaps and find strongly negative variance risk premiums. Furthermore, they consider different risk factors and find that none of them can explain the variance risk premiums. Variance risk premia are also considered by Bollerslev et al. (2011). In particular, they find that the extracted variance risk premium can in parts predict future asset returns.

A second approach for measuring model-free implied variance deals with similar but slightly different arguments. Bakshi et al. (2003) give prices and hedge portfolios for contracts that are derived from higher moments of the log-returns. In particular, they also consider variance contracts and additionally contracts for the skewness and kurtosis of the returns. They show that the market index is in general negatively skewed and individual stocks are typically less negatively skewed. In contrast to the variance swap rate their variance contract is not only based on the expectation of the log-contract but on the variance of the log-returns. A more general discussion of possible constructions for variance contracts is given in Bondarenko (2014). In contrast to our research question, the focus of these papers is on trading variance and specifications of variance contracts. We will study the differences between both model-free approaches. We show that the variance swap rate is just one part of log-return variances. Loosely speaking, only the quadratic variation from the diffusion component and the jump component are reflected in the variance swap rate. We show that volatility of stochastic variance, the correlation between stochastic variance and asset returns, and the higher moments from the jump distribution are neglected in the variance swap rate. Because the risk-neutral distribution is typically negatively skewed, we argue that this leads to an underestimation of the true variance of the returns under the pricing measure.

The rest of the paper is organized as follows. First, we will discuss theoretical foundations for both model-free variance measures and show what makes the difference between them. We argue that the second approach incorporates additional risk factors which are ignored in the market methodology. A simple example in a binomial setup is used to illustrate the difference. Subsequently, we perform an empirical analysis of the additional risk and calculate the difference between both measure from historical DAX-option prices. We show that it is indeed positive and non-constant. Using a regression model we show that

the difference can be almost completely explained by the variance swap rate. We argue that this means that additional sources of risk like stochastic correlation are ignored in financial markets. However, considering the difference has still some merits. We show that the difference can be used in efficient estimation methods for the parameters of option pricing models and can also be used for the purpose of model validation. We will illustrate the procedure using a Heston model.

2 Expected Variation and Implicit Variance

2.1 A general model setup

The following section illustrates the differences between the two measures of risk that builds the foundation of our research question. Therefore, we keep the model assumptions as general as possible. Nevertheless, for reasons of clarity, we refrain from a detailed treatment of the necessary mathematical prerequisites and we will also make some simplifying approximations. The reason for this is that the model only serves for illustrative purposes. Our intention is to show that the mere consideration of the variance swap rate is not sufficient. Various factors of model risk are in general neglected in the construction of variance indices like VIX or VDAX.

First of all, we denote by S_t the price of an asset and by $B_t(T)$ the price of a zero bond maturing in T at time t . The fundamental theorem of asset pricing implies the existence of an equivalent probability measure $\mathbb{Q}^T$ such that the asset price expressed in units of the numeraire $B(T)$ is a martingale. Assuming that the asset pays no dividends the discounted price process can also be interpreted as the price of a forward $F_t = \frac{S_t}{B_t(T)}$. No arbitrage then implies that there exists the so-called T -forward measure $\mathbb{Q}^T$ defined on a stochastic basis $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{Q}^T)$ such that the forward price process $(F_t)_{0 \leq t \leq T}$ is a martingale adapted to the filtration $\mathbb{F}$. From the representation theorem for martingales (see Jacod and Shiryaev (1987)) it is well-known that, under certain regularity conditions, every local martingale can be decomposed into purely continuous martingale and a purely discontinuous martingale, wherein the martingales are orthogonal to each other. Following Carr and Wu (2009), we start with the following representation for the forward price

$$dF_t = F_{t-} \left(\sigma_{t-} dW_t + \int_{\mathbb{R}} (e^x - 1) \tilde{J}(dx, dt) \right) \quad (2.1)$$

where F_{t-} denotes the forward price prior to any jump at time t , $(W_t)_{t \geq 0}$ is a $\mathbb{Q}^T$ -Brownian motion, and $\tilde{J}$ is a compensated jump measure. For clarity, we assume a

finite activity jump process. In this case (see, e.g., Cont and Tankov (2004)) we may write the compensated jump measure as $\tilde{J}(dx, dt) = J(dx, dt) - \nu_t(dx)dt$. Furthermore, the volatility σ_t and the jump compensator λ_t are in general assumed to be stochastic, predictable with respect to the filtration $\mathbb{F}$, and satisfy the usual hypotheses. A jump of size x as indicated by the random measure J then implies a jump of the forward price from F_{t-} to $F_t = F_{t-}e^x$.

Because F_t is a semimartingale we can apply Itô's formula. Itô's formula states that for any twice differentiable function f the following relationship holds:

$$f(F_t) - f(F_0) = \int_0^t f'(F_{s-})dF_s + \frac{1}{2} \int_0^t f''(F_{s-})d[F^c]_s + \sum_{0 \leq s \leq t} f(F_s) - f(F_{s-}) - \Delta F_s f'(F_{s-})$$

where F^c denotes the continuous part of F and $([F^c]_t)_{t \geq 0}$ its quadratic variation process. Applying the formula to Equation 2.1 and the function $f(F) = \ln F$, it follows

$$\begin{aligned} \ln \frac{F_t}{F_0} &= \int_0^t \sigma_{s-} dW_s + \int_0^t \int_{\mathbb{R}} (e^x - 1) \tilde{J}(dx, ds) \\ &\quad - \frac{1}{2} \int_0^t \sigma_{s-}^2 ds + \int_0^t \int_{\mathbb{R}} (x - e^x + 1) J(dx, ds) \end{aligned} \quad (2.2)$$

Since $e^x - 1 \approx x$, we further have

$$\approx \int_0^t \sigma_{s-} dW_s + \int_0^t \int_{\mathbb{R}} x \tilde{J}(dx, ds) - \frac{1}{2} \int_0^t \sigma_{s-}^2 ds \quad (2.3)$$

Both, the approximation and the true solution, result in the equation for the quadratic variation of the forward returns until T

$$[\ln F]_T = \int_0^T \sigma_{s-}^2 ds + \int_0^T \int_{\mathbb{R}} x^2 J(dx, ds) \quad . \quad (2.4)$$

such that the risk-neutral expectation of the variation of future returns is given by

$$\text{SW}(T) = \mathbb{E}^T [[\ln F]_T] = \mathbb{E}^T \left[\int_0^T \sigma_{s-}^2 ds + \int_0^T \int_{\mathbb{R}} x^2 J(dx, ds) \right]$$

where SW denotes the variance swap rate along the lines of Carr and Wu (2009).

In the remainder of the paper we are mainly interested in the first two moments of the forward returns. Assuming that all occurring integrals are well-defined and in particular that integrands of stochastic integrals are predictable, square-integrable processes, we have

Proposition 2.1 Denote the integrated (stochastic) variance process by $V_t = \int_0^t \sigma_s^2 ds$. The first two moments under the risk-neutral measure $\mathbb{Q}^T$ of the approximated forward returns in Equation 2.3 until T are given by

$$\begin{aligned} M_1(T) &:= \mathbb{E}^T \left[\ln \frac{F_T}{F_0} \right] = -\frac{1}{2} \mathbb{E}^T [V_T] \\ M_2(T) &:= \mathbb{E}^T \left[\left(\ln \frac{F_T}{F_0} \right)^2 \right] \\ &= \mathbb{E}^T [V_T] + \mathbb{E}^T \left[\int_0^T \int_{\mathbb{R}} x^2 J(dx, ds) \right] + \frac{1}{4} \mathbb{E}^T [V_T^2] \\ &\quad - \left(\text{Cov}^T \left(V_T, \int_0^T \sigma_s ds W_s \right) + \text{Cov}^T \left(V_T, \int_0^T \int_{\mathbb{R}} x \tilde{J}(dx, ds) \right) \right) \end{aligned}$$

The derivation is given in the Appendix.³ Observe that the difference A between the second moment of forward returns and the variance swap rate is given by three terms

$$\begin{aligned} A(T) &:= M_2(T) - SW(T) \\ &= \frac{1}{4} \mathbb{E}^T [V_T^2] - \text{Cov}^T \left(V_T, \int_0^T \sigma_s ds W_s \right) - \text{Cov}^T \left(V_T, \int_0^T \int_{\mathbb{R}} x \tilde{J}(dx, ds) \right) . \end{aligned}$$

Alternatively, we may consider the difference between the variance of the returns and the expected variation. It holds

$$\begin{aligned} B(T) &:= \text{Var}^T \left[\ln \frac{F_T}{F_0} \right] - SW(T) = M_2(T) - M_1(T)^2 - SW(T) \\ &= \frac{1}{4} \text{Var}^T [V_T] - \text{Cov}^T \left(V_T, \int_0^T \sigma_s ds W_s \right) - \text{Cov}^T \left(V_T, \int_0^T \int_{\mathbb{R}} x \tilde{J}(dx, ds) \right) . \end{aligned}$$

While the variance swap rate only measures the diffusion and jump risk, additional factors are reflected in the second moment and the variance of the forward returns. In the approximated model, these risks include

- The variance of the integrated variance
- The correlation between the integrated variance and the diffusion part of the forward prices.

³Without the approximation, the first moment is given by

$$M_1(T) = -\frac{1}{2} \mathbb{E}^T [V_T] + \mathbb{E}^T \left[\int_0^T \int_{\mathbb{R}} (x - e^x + 1) J(dx, ds) \right] .$$

The second moment contains additional terms resulting from the jump distribution.

- The correlation between the integrated variance and the jump part of the forward prices.

For the purpose of model risk, it is necessary to consider this additional factors explicitly. They have an outstanding importance for optimal portfolio strategies, hedging strategies, and may also give deeper insights of the uncertainty about the price processes.

Before we consider the model-free setup, we discuss some special model assumptions. In particular, we exclude jumps in all three cases. Observe, that the approximation above only concerns the jump part, such that the reasoning is exact for these special cases.

No jumps In this case the variance swap rate is determined by the first moment of the returns.

$$SW(T) = \mathbb{E}^T [\ln F]_T = -2\mathbb{E}^T \left[\ln \frac{F_T}{F_0} \right],$$

which is the foundation of the classic result of Britten-Jones and Neuberger (2000) and its extension by Jiang and Tian (2005). Furthermore, the additional risk measured by the second moment reflects the variance of integrated (stochastic) variance and the covariance between forward price process and the (stochastic) variance process.

$$A(T) = \frac{1}{4}\mathbb{E}^T [V_T^2] - \text{Cov}^T \left(V_T, \int_0^T \frac{dF_s}{F_s} \right)$$

The additional risk can take both positive and negative values depending on the correlation between stochastic variance and the forward price. In particular, negative values for the additional factor immediately imply a positive correlation. Positive values may correspond to a negative correlation or a positive correlation which is offset by a large variance of integrated variance.

No jumps, uncorrelated variance Additionally, if the forward price process is uncorrelated with stochastic volatility (or variance) it follows

$$\text{Var}^T \left[\ln \frac{F_T}{F_0} \right] = \mathbb{E}^T [\ln F]_T + \frac{1}{4} \text{Var}^T [V_T]$$

or

$$B(T) = \text{Var}^T \left[\frac{1}{2} V_T \right]$$

Return variance is given by the variance swap rate and variance of integrated variance. In particular, the difference between both variance measures will always

be positive. Conversely, the variance of the integrated variance can be determined from the difference between the variance swap rate and the return variance.

No Jumps, deterministic volatility In case of deterministic volatility the expected variation and return variance coincide.

$$\text{Var}^T \left[\frac{\ln F_T}{F_0} \right] = \mathbb{E}^T [[\ln F]_T]$$

Therefore, a necessary condition for deterministic volatility and no jumps in the financial market is given by

$$B(T) = 0 \Leftrightarrow A(T) = M_1(T)^2$$

2.2 Model-free setup

Based on the work of Breeden and Litzenberger (1978) or Carr and Madan (2001) it is possible to calculate the moments M_1 and M_2 in a model-free manner from a continuum of option prices. Interpreting the moments as payoffs from contingent claims, it can be spanned and priced using explicit positioning across option strikes. Because option prices reflect risk-neutral expectations, the prices of the portfolios equals the moments under the martingale measure. Furthermore, the spanning result also gives an explicit construction of static portfolios consisting of out-of-the-money calls and puts

Proposition 2.2 *The model-free first two moments under the measure $\mathbb{Q}^T$ of the forward returns until T are given by*

$$M_1(T) = -\frac{1}{B_0(T)} \left(\int_0^{F_0} \frac{\text{Put}_0(S_0, K, T)}{K^2} dK + \int_{F_0}^{\infty} \frac{\text{Call}_0(S_0, K, T)}{K^2} dK \right) \quad (2.5)$$

$$\begin{aligned} M_2(T) = \frac{2}{B_0(T)} & \left(\int_0^{F_0} \frac{\text{Put}_0(S_0, K, T)}{K^2} dK + \int_{F_0}^{\infty} \frac{\text{Call}_0(S_0, K, T)}{K^2} dK \right) \quad (2.6) \\ & - \frac{2}{B_0(T)} \left(\int_0^{F_0} \frac{\left(\ln \frac{K}{F_0} \right) \text{Put}_0(S_0, K, T)}{K^2} dK \right. \\ & \quad \left. + \int_{F_0}^{\infty} \frac{\left(\ln \frac{K}{F_0} \right) \text{Call}_0(S_0, K, T)}{K^2} dK \right) \end{aligned}$$

where $\text{Put}(S_0, K, T)$ and $\text{Call}(S_0, K, T)$ denote prices for European puts and calls for asset prices S_0 , strikes K , and maturity T .

The proof is given in the Appendix. Similar considerations are made in Bakshi et al. (2003) and Guillaume and Schoutens (2013a). In particular, $M_2(T)$ is the expected payoff of the variance contract in the sense of Bakshi et al. (2003). As we have shown in Subsection 2.1 it is necessary to distinguish this contract from the variance indices like VIX and VDAX or the variance swap rate. These contracts are based on the expected variation. For completeness we replicate the results of the literature for the variance swap rate. Carr and Wu (2009) show that in the model setup of Subsection 2.1 the expected variation can be expressed as

$$\mathbb{E}_t^T [[\ln F]_T] = -2M_1(T) + \epsilon$$

where ϵ denotes an approximation error, which is zero when the asset price process is purely continuous and of order $O\left(\left(\frac{dF_t}{F_{t-}}\right)^3\right)$ when forward prices can jump. Typically, in actual variance contracts the approximation error is ignored and the variance swap rate is given by

$$SW(T) = -2M_1(T) \quad (2.7)$$

We call this method for calculating model-free implied variances the market methodology, because it is also used in volatility indices like VIX. Let us once again emphasize, that it is based on the expected variation of the returns and is based only on the first moment of the returns. Possible misjudgments resulting from this measure are considered in the next subsection.

Beforehand, we present some simplifications of Equation 2.5 and 2.6. If market prices are expressed in implied volatilities for the Black-Scholes model and one defines forward moneyness by $m = \frac{F_0}{K}$, a simple application of the substitution rule for integration gives the alternative expression

$$\begin{aligned} M_1(T) &= - \left(\int_0^1 \frac{\text{Put}_0(m, T)}{m^2} dm + \int_1^\infty \frac{\text{Call}_0(m, T)}{m^2} dm \right) \\ &= - \int_0^\infty \frac{\Theta_0(m, T)}{m^2} dm \end{aligned} \quad (2.8)$$

$$\begin{aligned} M_2(T) &= 2 \left(\int_0^1 \frac{\text{Put}_0(m, T)}{m^2} dm + \int_1^\infty \frac{\text{Call}_0(m, T)}{m^2} dm \right. \\ &\quad \left. - \int_0^1 \frac{(\ln m) \text{Put}_0(m, T)}{m^2} dm + \int_1^\infty \frac{(\ln m) \text{Call}_0(m, T)}{m^2} dm \right) \\ &= 2 \left(\int_0^\infty \frac{\Theta_0(m, T)}{m^2} dm - \int_0^\infty \frac{(\ln m) \Theta_0(m, T)}{m^2} dm \right) \\ &= SW(T) + A(T) \end{aligned} \quad (2.9)$$

with

$$A(T) := -2 \int_0^\infty \frac{(\ln m) \Theta(m)}{m^2} dm. \quad (2.10)$$

Here, $\text{Put}_t(m, T)$ and $\text{Call}_t(m, T)$ denote prices of puts and calls in the Black-Scholes model with maturity T and moneyness m at time t where the risk-free interest rate is zero. Simplifying the notation, Θ denotes the price of a put option if the moneyness is less than 1 and the price of a call option if $m > 1$.

Using this representation, we do not need any information about the risk-free rate or zerobond prices. The moneyness can be calculated from the forward price. The forward price itself can be derived from the put-call parity. The forward price equals the strike for which put and call prices are equal.

Interestingly, but not surprisingly, in the model-free setup we have an identical decomposition of the second moment as in the general model. We conclude, that the second integral reflects the risk from the variances and covariances and the first integral reflects the diffusion and jump risk.

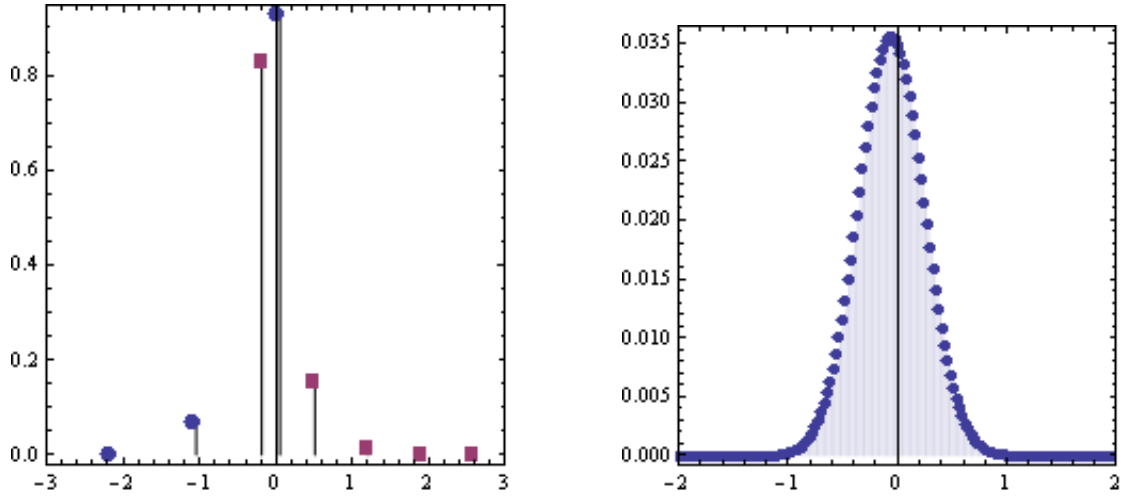
2.3 Example: Binomial trees

In order to obtain a better understanding of the differences between variance swap rates and the return variances, we now consider a binomial model.

An advantage of binomial trees over the Black-Scholes model is an additional degree of freedom in the choice of parameters. It is well-known, that the binomial distribution can approach both to the normal distribution and the poisson distribution, depending on how the parameters are chosen. We can make use of this in order to generate skewed and kurtotic distributions for the return process and discuss the impact on the two model-free implied variance measures. Therefore, we fix the variance of the returns and look at the factors $A(T)$ and $B(T)$, which were introduced in Subsection 2.1.

In each period, up movement in a binomial model are denoted a percentage increase $u - 1$ of the asset price. The percentage decrease for down movements is $1 - d$. The price of the underlying at time S_{i+1} simply moves up to uS_i or down to dS_i . Assuming a risk-free interest rate of r per period, the risk-neutral probability of an up movement is given by $q = \frac{1+r-d}{u-d}$ and of a down movement by $1 - q$. For the (central) moments of

Figure 1: Return probabilities for different binomial trees



Left side: Circled markers give the probabilities of the log-returns for scenario I. The squares give probabilities for scenario III. Right side: Distribution for scenario II.

the N -period forward returns it holds

$$M_1^{\text{bin}}(t_N) = N \times \left(q \ln \frac{u}{1+r} + (1-q) \ln \frac{d}{1+r} \right)$$

$$\text{Var}^{T,\text{bin}} \left[\ln \frac{F_{t_N}}{F_0} \right] = M_2^{\text{bin}}(t_N) - (M_1^{\text{bin}}(t_N))^2 = N \times q \times (1-q) \left(\ln \frac{u}{d} \right)^2$$

We consider three different cases in Table 1. For all three cases we assume a risk-free rate equal to zero, an annualized standard deviation of the (log-)returns $\sigma = 30\%$, and a number of periods $N = 500$. The scenarios I and III corresponds to rare events. Scenario I describes rare negative jumps and III describes rare positive jumps. Scenario II can be interpreted as an approximation of the Black-Scholes model with a symmetric distribution of gains and losses. The log-return distributions are illustrated in Figure 1.

To gauge how differences between market methodology and return variances are affected skewness and kurtosis of returns, we chose two rather extreme scenarios I und III. As mentioned, scenario II can be viewed as a benchmark similar to a Black-Scholes model. From Table 1 we conclude that in the left skewed model (scenario I) the market methodology underestimates return variance and in the right skewed model (scenario III) it overestimates return variance. Interpreting the model-free implied variances as risk measures this could be problematic. In general a risk measure is more concerned with

Table 1: Characteristics for different binomial models with equal return variances

Scenario	$\{d, u\}$	q	Mean	Std. dev.	Skewness	Kurtosis
I	$\{0.3348, 1.0001\}$	0.9998	-0.03225	0.30	-3.6470	16.304
II	$\{0.9862, 1.0130\}$	0.5155	-0.04499	0.30	-0.0028	2.996
III	$\{0.9996, 2.000\}$	0.0004	-0.05748	0.30	2.3100	8.332

Scenario	$\sqrt{\text{SW}(t_N)}$	$A(t_N)$	$B(t_N)$
I	25.3965%	0.02654	0.02550
II	29.9954%	0.00205	0.00003
III	33.9043%	-0.0216	-0.0250

The upper part of the table gives the characteristics of the binomial tree. The first 3 columns contain the model parameters of the respective scenarios. Columns 4–7 give the central moments of the (log) returns.

The lower part shows the differences between using market methodology and return variances. Columns 2 denotes the model-free implied volatility resulting from VIX methodology. Columns 3 and 4 give the differences between variance swap rate and second moment and return variances.

downward movements and we typically observe left skewed return distributions. Using the variance swap rate as a proxy for return variance, it would underestimate the true return variance (0.09) by more than 25%. In contrast, for an approximately symmetric distribution like scenario II, we see that differences are rather small. But note, that kurtosis of scenario II is also quite similar to normal distribution kurtosis. The effect of a possibly leptokurtic distribution is beyond the scope of this simple binomial example.

3 Empirical Analysis

3.1 Data

The data used in this study is from the Bloomberg Database. The data sample starts in January 2006 and lasts till August 2011. We use daily closing prices for options on the Deutscher Aktienindex (DAX). The DAX measures the performance of the 30 largest German Companies. As a performance index, dividends are reinvested and can be ignored for the rest of the paper. According to the calculation described in Subsection 2.2 we use at- and out-of-the money options. Options with zero-bid prices are excluded from

the analysis. For options with non-zero-bid prices we calculate for each trading day the mid prices from their last bid and ask prices. For the analysis itself, we use the implied volatilities which are also given by the Bloomberg Database.

For the analysis we concentrate on 3 and 6-month horizons for the variance measures. We exclude shorter horizons because we want to exclude possible pricing anomalies that can occur for options close to expiration. Longer horizons are excluded because of two reasons. First, in general options with longer maturities are less liquid and we suspect results will be less meaningful. Second, the relevant area of the implied risk-neutral distribution is broader for longer horizons. This means, that we would need additional strikes for our analysis. Unfortunately we typically observe option prices for moneyness regions between 0.8 and 1.2 only. We suspect that results for longer horizons will be more sensitive to the interpolation methods and therefore exclude them in this analysis. To get constant 3- and 6-month horizons, we first linearly interpolate implied volatilities from the the two nearest maturities. Secondly, we interpolate implied volatilities with respect to the strikes. For strikes below and above the lowest and highest available strikes we use the implied volatility at the lowest and highest strike, respectively. Because we make use of the moneyness representations in Equation 2.8 and Equation 2.9 we do not need information about the risk-free interest rate. A discussion which interest rate could be interpreted as risk-free can be ignored. To calculate forward prices, we use strikes for which the price of the call and the put are equal. The forward price is then set to this strike and we calculate the moneyness for the options in our dataset and their prices from the Black-Scholes formula.

Finally, we compute variance swap rates SW and the difference A between variance swap rates and second moments of the returns by numerical integration techniques. For the two different maturities the respective values are annualized by dividing with time horizons $\frac{1}{4}$ and $\frac{1}{2}$.⁴

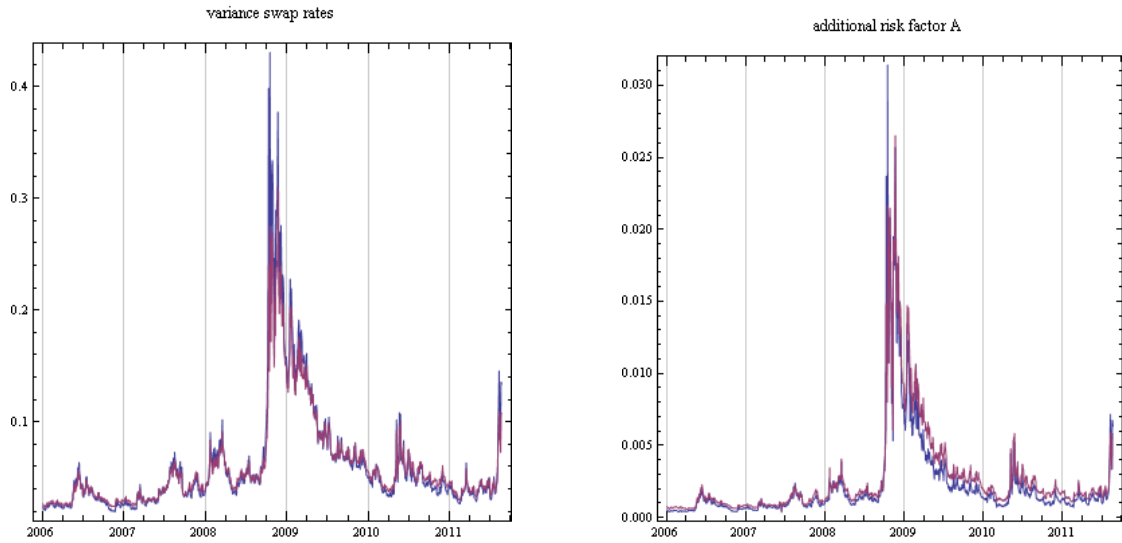
The resulting time series are illustrated in Figure 2.

3.2 Results

In line with the example for the Binomial model we see that the difference is in general rather small. Furthermore, at a first glance, time series for the variance swap rate series and for the additional component seem to possess similar characteristics. This is no surprise because we have already shown in Subsection 2.1 that the additional risk

⁴Additionally to A we also consider the difference B between the variance of the log-return and the variance swap rate. The results are nearly identical and so we do not report them here.

Figure 2: Time series for variance swap rate and the additional risk factor A



Left chart illustrates the history of the normalized 3- and 6-month variance swap rate from 2006 to 2011. Right chart illustrates the history of the additional risk factor A introduced in Section 2. Blue lines represent 3-month swap rates. Red lines represent 6-month swap rates.

factor contains the variance of the integrated variance process and its covariance with the underlying. Assuming that these are proportional to the instantaneous variance one would expect this pattern. For example, such proportionality is implied by the Heston model. Therefore, we also consider the ratio of the additional risk and the variance swap rate in Figure 3. Besides the observation that the ratio still takes its maximum at the end of 2008, we observe that the ratio appears to be slightly larger in the years following the crisis. In particular, this is even more pronounced for the 6-month horizon. This could be an indication of a (small) regime switch in the years following the financial crisis. We also observe that the variance swap rate itself reverts more or less to the level before the crisis.

Summary statistics for the six different time series are given in Table 2. Additionally, we also consider the increments of the respective time series. We report these values because econometric modeling is typically more concerned with the increments and the increments could, for example, be assumed to be a stationary process (see, e.g., Alexander (2008, chap. II.5)). We see that for 3- and 6-month horizons the results only differ marginally. All distributions are right-skewed and have a high kurtosis. Our main focus is on the properties of the additional risk factor. As we will see in the following

Figure 3: Ratio of additional risk factor A to variance swap rate

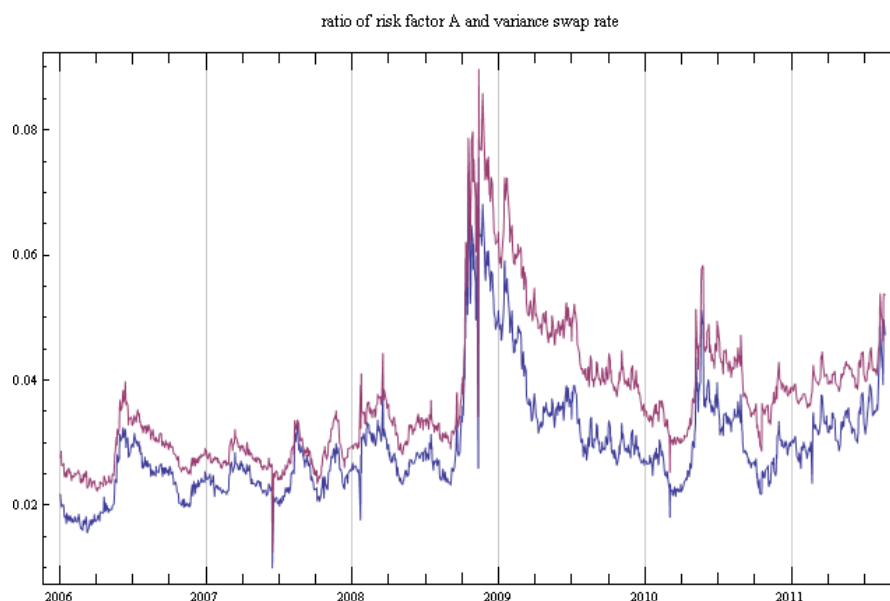


Illustration of the history of the ration of the additional risk factor A to the variance swap rate for 3- and 6-month horizons from 2006 to 2011. Blue lines represent 3-month swap rates. Red lines represent 6-month swap rates.

regression analysis, this factor can be explained to a large extent by the variance swap rate. Therefore we will refrain from further econometric analysis of the underlying processes. Detailed discussions of the variance swap rate or VIX and its relations to realized variations can be found in Jiang and Tian (2005), Carr and Wu (2009), Becker et al. (2009), Duan and Yeh (2010), and Bollerslev et al. (2011).

We want to shed light on the question is whether the additional term $A(T)$ possesses an independent source of risk or if it can be explained by variance swap rates. An independent source of risk would imply that the market methodology indeed ignores certain risk factors. Potentially, this risk factors could have explanatory power for realized variances. Based on the considerations in Subsection 2.1 additional risk factor(s) would imply that either variance-of-variance, the correlation between the diffusion processes, or the correlation between jump and diffusion processes are stochastic processes with independent risk factors. A functional relation between variance swap rates and the additional terms would suggest that a two-factor model – stochastic asset returns with stochastic volatil-

Table 2: Summary Statistics for variance swap rate, additional term A , and ratio

Series	Daily values $\times 100$				Perc. Differences in daily values			
	Mean	Std.dev.	Skew	Kurt	Mean	Std.dev.	Skew	Kurt
3M-SW	6.17	5.18	3.02	10.88	0.49	8.94	1.63	9.84
$A(3M)$	0.22	0.32	4.09	20.10	1.09	15.37	5.41	73.23
3M-Ratio	2.97	0.88	1.71	3.97	0.28	7.88	12.07	276.40
6M-SW	6.04	4.22	2.56	7.23	0.32	6.85	1.57	9.72
$A(6M)$	0.27	0.32	3.41	13.26	0.74	12.23	5.57	79.84
6M-Ratio	3.73	1.11	1.52	2.77	0.21	6.51	12.90	311.36

Table reports summary statistics for the variance swap rate, the additional risk factor A , and the ratio of the risk factor to the variance swap rate for 3 and 6 months horizons. The statistics are given for the daily values and for the percentage differences in daily values. Columns Mean, Std.dev., Skew, and Kurt report the sample average, standard deviation, skewness, and excess kurtosis, respectively.

ities – is sufficient. A first hint on the relation is given in Figure 4, where we observe that a functional relation seems to exist at least for small values of the variance swap rate.

Therefore, we consider two regression models:

$$A(T) = \beta_2 \text{SW}(T) + \epsilon \quad (3.1)$$

$$A(T) = \beta_1 \sqrt{\text{SW}(T)} + \beta_2 \text{SW}(T) + \beta_3 \text{SW}(T)^2 + \epsilon \quad (3.2)$$

The coefficient estimates are given in Table 3.

We see that most of the variation in the additional risk factor can be explained by the variance swap rate. For the assumed relations we observe an adjusted R-squared of up 99%. Therefore, we can conclude that correlation and variance-of-variance are deterministic functions of already considered risk factors. In a specific model it is sufficient to specify this relation correctly and the specification can be extracted from historic values. The daily calibration procedure for a specific model of option prices should refrain from recalibrations of the scale parameters. Instead only location parameter must be recalibrated. We will make use of this observation in the next section. But, it must be mentioned, that in the case of market turmoil (when variance swap rates are large) we do see unexplainable differences. In these extrem situations, the assumption of no independent risk factors might be wrong.

Figure 4: Dependence of additional risk factor and variance swap rate

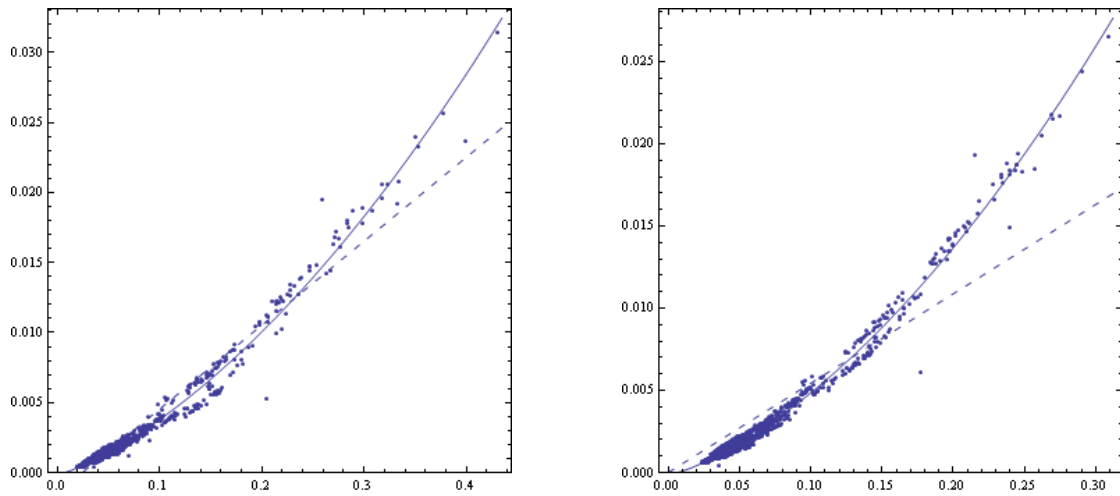


Illustration of the dependence of the additional risk factor and the variance swap rate. Left and right figure shows the dependence for 3-month and 6-month horizons, respectively. The solid and dashed lines represents the relation resulting from regression models 3.1 and 3.2, respectively.

4 Moment-based Parameter Estimation: The Heston model

In the following section we would like to show how the difference between the measures can be utilized for estimating model parameters. We illustrate the procedure for the Heston model. But, similar arguments can be applied for any model, where the characteristic function of the log-returns are known. In particular, the characteristic function is typically known in most Lévy models. For an overview of characteristic functions for different models we refer to Cont and Tankov (2004).

We would like to mention, that similar arguments are made in Guillaume and Schoutens (2013a). The arguments here differ in two ways. First, we estimate the parameters from the time series considered in Section 3. In contrast, the arguments in Guillaume and Schoutens (2013a) are used for daily calibrations of the model. More significant is the second distinction. Guillaume and Schoutens (2013a) explicitly ignore the first moment of the log-return and only consider higher moments. As we have argued, it is the first moment that describes the variance swap rate and that it explains much of the variation in the second moment and the variation in the difference between the second moment and the variance swap rate. Using the functional relation means that recalibrations of

Table 3: Regression results

Reg	β_1	β_2	β_3	R^2	AIC
3 month horizon					
(I)		0.0460 (121)		0.9105	-15 317
(II)	-0.0033 (-13.6)	0.0390 (34.4)	0.0937 (36.1)	0.9895	-18 383
6 month horizon					
(I)		0.0544 (122)		0.9117	-15 106
(II)	-0.0035 (-12.2)	0.0430 (29.8)	0.1664 (40.8)	0.9933	-18 791

Entries report the results for the two regression models given in Equation 3.1 (I) and Equation 3.2 (II) for 3- and 6-month horizons. The t -statistics for the coefficients are in parentheses. The column R^2 reports the adjusted R-squared and AIC reports the Akaike information criterion.

the scale parameters are unnecessary. For the parameter estimation, our method makes explicit use of the first two moments.

We will start with a brief overview of the relevant results for the Heston model. In the following we will describe a potential algorithm for the parameter estimation.

4.1 The Heston model

The Heston model is given by the following stochastic differential equations

$$dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t^1 \quad (4.1)$$

$$dF_t = F_t\sqrt{v_t}dW_t^2 \quad (4.2)$$

where (W^1, W^2) is a two-dimensional Brownian motion (possibly correlated) under the risk-neutral measure P^* . Here, v_t is the variance process and F_t is the forward price process. Applying Itô's formula with $x_t = \ln \frac{F_t}{F_0}$, we get

$$dx_t = -\frac{1}{2}v_t dt + \sqrt{v_t}dW_t^2.$$

The characteristic function of x_t is well-known in financial literature. One representation is based on Schoutens et al. (2004) or Gatheral (2006), which reads

$$\begin{aligned}\phi_x(u, t) &:= \mathbb{E}_0 [\exp(i u x_t)] \\ &= \exp \left(\frac{\kappa \theta}{\sigma^2} \left((\kappa - i \rho \sigma u - d)t - 2 \ln \frac{1 - g e^{-dt}}{1 - g} \right) \right) \\ &\quad \times \exp \left(\frac{v_0(\kappa - i \rho \sigma u - d)(1 - e^{-dt})}{\sigma^2(1 - g e^{-dt})} \right)\end{aligned}$$

where

$$d = \sqrt{(i \rho \sigma u - \kappa)^2 + \sigma^2(i u + u^2)} \text{ and } g = \frac{\kappa - i \rho \sigma u - d}{\kappa - i \rho \sigma u + d}.$$

The characteristic function can be used to calculate moments of x_t . Differentiating n times we get

$$\mathbb{E}_0 [x_t^n] = i^{-n} \left[\frac{\partial^n \phi_x(u, t)}{\partial u^n} \right]_{u=0}.$$

For the first two moments we get for the Heston model

$$M_1^H(t) = \mathbb{E} [x_t] = -\frac{1}{2} \left(\theta t + \frac{(v_0 - \theta)(1 - e^{-\kappa t})}{\kappa} \right) \quad (4.3)$$

$$\begin{aligned}M_2^H(t) = \mathbb{E} [x_t^2] &= \theta t + \frac{(v_0 - \theta)(1 - e^{-\kappa t})}{\kappa} + \left(\frac{(v_0 - \theta)(1 - e^{-\kappa t}) + \kappa \theta t}{2\kappa} \right)^2 \\ &\quad + \frac{(v_0 - \theta)(e^{-\kappa t} - 1 + \kappa t e^{-\kappa t}) - \theta(\kappa t + e^{-\kappa t} - 1)}{\kappa^2} \rho \sigma \\ &\quad + \frac{(v_0 - \theta)(2 - 4\kappa t e^{-\kappa t} - 2e^{-2\kappa t}) - \theta(3 - 2\kappa t - 4e^{-\kappa t} + e^{-2\kappa t})}{(2\kappa)^3} \sigma^2\end{aligned} \quad (4.4)$$

Using the general relation between the expected variation and the first moment of the return presented in Subsection 2.1 for a process with no jumps we have⁵

$$\text{SW}_0^H(t) = \mathbb{E} [V_t] = \theta t + \frac{(v_0 - \theta)(1 - e^{-\kappa t})}{\kappa}. \quad (4.5)$$

Note, that in the Heston model the differences A and B have the simple representations

$$\begin{aligned}A_0^H(t) &= B_0^H(t) + \frac{1}{4} \text{SW}_0^H(t)^2 \\ B_0^H(t) &= \frac{(v_0 - \theta)(e^{-\kappa t} - 1 + \kappa t e^{-\kappa t}) - \theta(\kappa t + e^{-\kappa t} - 1)}{\kappa^2} \rho \sigma \\ &\quad + \frac{(v_0 - \theta)(2 - 4\kappa t e^{-\kappa t} - 2e^{-2\kappa t}) - \theta(3 - 2\kappa t - 4e^{-\kappa t} + e^{-2\kappa t})}{(2\kappa)^3} \sigma^2.\end{aligned}$$

⁵Alternatively, the expected integrated variance can be calculated explicitly from formulas given by Cox et al. (1985) or Dufresne (2001). Of course, both approaches give the same results.

As expected, we observe that the first moment and the variance swap rate only depend on the drift parameters κ and θ of the variance process. Volatility σ of the variance process and the correlation ρ between the variance process and the forward price process are purely reflected by the second moment. For calibration and estimation procedures it seems to make sense, to make use of this separation.

4.2 Risk-neutral Parameter Estimation Heston Model

The main problem for estimation procedures of the Heston Model is the unobservable spot variance v_t at each date $t \in [0, T]$ in Equation 4.1 and Equation 4.2. Typically the literature (e.g., Bakshi et al. (1997)) deals with it in the following way. For each day the spot variance and the structural parameters are backed out from option prices observed at that day. Therefore an objective function, which defines the pricing error, is minimized with respect to the spot variance and the structural parameters. An estimation of the structural parameters is then simply given by the mean of daily calibrated parameters. As Guillaume and Schoutens (2013b) point out the solution of the daily optimization procedure might be instable with respect to small changes in option prices and therefore implies calibration risk. Furthermore, the objective function is typically a nonconvex function of the parameters and can thus have several minima.

We propose an alternative methodology that divides the single optimization with respect to all parameters at the same time in a multi-step procedure. First, we single out the spot variance via its relation to the variance swap rate. Then, variance swap rates are used for estimating the location parameters of the stochastic variance process. Finally, the scale parameters are estimated. Therefore we use the empirical differences A or B and calibrate the model differences. Singling out the spot variance has an additional advantage. Having determined all model parameters we can inversely calculate the implicit daily spot variances corresponding to the model. Considering the times series of implicit daily spot variances then gives us the opportunity to validate the model.

The considerations above imply two simple least-squares optimization problems for the estimation of the Heston parameters. The algorithm works as follows

- Calculate empirical variance swap rates $\widetilde{SW}_i(T)$ for each sample date $i = 1, \dots, M$ and each option maturity $T \in \{T_1, \dots, T_n\}$. The empirical variance swap rates can be calculated from the representations in Equation 2.7 and Equation 2.8.
- Calculate the empirical differences $\tilde{A}_i(T)$ or $\tilde{B}_i(T)$ for each date i and each maturity T . The empirical differences can be calculated from Equation 2.10.

- To single out the spot variance we solve Equation 4.5 for $(v_t - \theta)$

$$(v_t - \theta) = \frac{(\text{SW}_t(T) - \theta(T - t)) \kappa}{1 - e^{-\kappa(T-t)}} =: \xi_t(T) \quad (4.6)$$

This quantity should not depend on variance swap maturities. Using empirical variance swap rates, this leads to the first objective function

$$\min_{\theta, \kappa} \sum_{i=1}^M \sum_{(T_1, T_2) \in \mathcal{T}} \left(\tilde{\xi}_i(T_1 | \theta, \kappa) - \tilde{\xi}_i(T_2 | \theta, \kappa) \right)^2 \quad (4.7)$$

where $\mathcal{T}$ is the 2-combination of the set containing the option maturities $T_1, \dots, T_n$. $\tilde{\xi}_i$ are the observations resulting from empirical swap rates in Equation 4.6 for each pair (θ, κ) . The objective function is the sum of squared errors between any two option maturities.⁶

- Based on Equation 4.6 we can also single out the spot variance in the differences A and B . Using the optimal parameters $(\hat{\kappa}, \hat{\theta})$ from optimization problem 4.7 we have

$$B_t^H(T | \rho, \sigma) = \frac{\xi_t(T)(e^{-\hat{\kappa}\tilde{T}} - 1 + \hat{\kappa}\tilde{T}e^{-\hat{\kappa}\tilde{T}}) - \hat{\theta}(\hat{\kappa}\tilde{T} + e^{-\hat{\kappa}\tilde{T}} - 1)}{\hat{\kappa}^2} \rho \sigma + \frac{\xi_t(T)(2 - 4\hat{\kappa}\tilde{T}e^{-\hat{\kappa}\tilde{T}} - 2e^{-2\hat{\kappa}\tilde{T}}) - \hat{\theta}(3 - 2\hat{\kappa}\tilde{T} - 4e^{-\hat{\kappa}\tilde{T}} + e^{-2\hat{\kappa}\tilde{T}})}{(2\hat{\kappa})^3} \sigma^2$$

where $\tilde{T} := T - t$. Using the empirical differences $\tilde{B}$ we identify the second optimization problem as follows

$$\min_{\rho, \sigma} \sum_{i=1}^M \sum_{j=1}^n \left(\tilde{B}_i(T) - B_i^H(T | \rho, \sigma) \right)^2 \quad (4.8)$$

We give results of the estimation procedure in Table 4. The estimates are given for the total sample from January 2006 – August 2011 and two subsamples from January 2006 – September 2008 and October 2008 – August 2011. According to Figure 2 we may interpret the first subsample as low-variance regime. The large variance-spikes occur in the second sample. Basically we find parameters that are similar to parameters reported in the literature⁷. Two observations are worth mentioning. First, in all samples the

⁶Of course, other objective functions could be considered (e.g., percentage errors). But, our results seem to stable with respect to different characterizations.

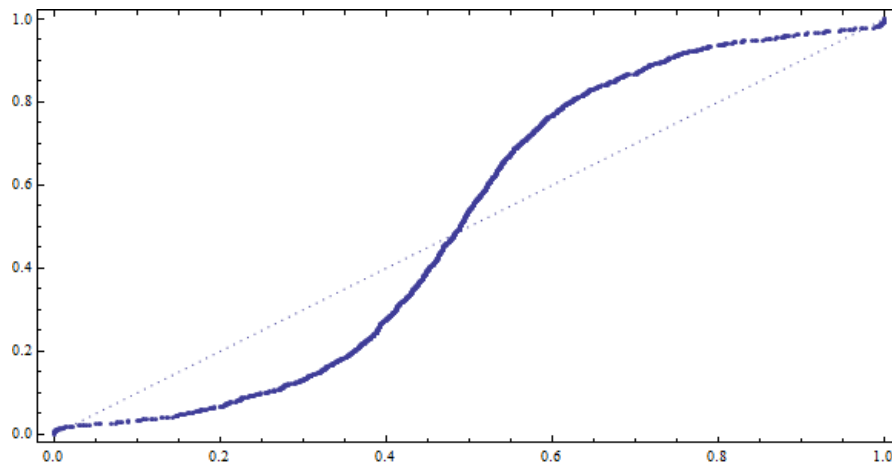
⁷For example Bakshi et al. (1997) give estimates for option-based calibration. Eraker et al. (2003) uses Markov Chain Monte Carlo methods for a likelihood-based estimation from return data.

Table 4: Parameter Estimation Heston Model

	θ	κ	σ	ρ
(I)	0.0820	0.3049	0.1522	-1
(II)	0.0643	0.4037	0.1474	-1
(III)	0.0864	0.4788	0.1860	-1

(Annualized) Parameter estimation for the Heston model based on the total sample (I) of option prices from January 2006 – August 2011 and two subsamples (January 2006–September 2008 (II) and October 2008 – August 2011 (III))

Figure 5: QQ-plot for implied spot variances



optimized correlation achieves its smallest possible value -1. Second, as expected from the discussion in 3, the estimates for the scale parameters are more stable than for location parameters. Nevertheless, interpretation of the results is difficult. It is well-known that the Heston model is misspecified. Therefore we present the QQ-plot for the increments of the implied spot variances. These are extracted from variance swap rates after the parameter estimation according to Equation 4.6. In the Heston model we would expect that these are approximately normal. In contrast, we observe strong deviations from normal probability. In particular, we conclude higher kurtosis. In future research we would like to test the algorithm in more meaningful models. Here, the main task is simply to illustrate a potential algorithm.

5 Conclusion

We analyzed the difference between two measures of model-free implied variance. The difference results from additional factors that influence the return variance and that are not embodied in the variance swap rate. In general this may pose problems if the variance swap rate is used for asset allocation or hedging purposes. Nevertheless, we also showed that the difference can be explained by the variance swap rate. This implies that well-specified model for stochastic variance and stochastic returns may be sufficient for this tasks. Furthermore, we showed that one can make use of the difference for parameter estimation. Especially for the scale parameter we get robust results even in a typically misspecified model.

It would be interesting to more closely compare the information content of both measures. Especially it would be interesting if measures for realized variances can be decomposed in similar ways. This may lead to further insights concerning the variance risk premium. Furthermore, applications to optimal portfolio strategies might also be interesting. It is well-known that for investors with finite horizons the intertemporal hedging demand strongly depends on the correlation between asset returns and stochastic volatility. Our approach may provide additional insights. Lastly, parameter estimates based on the proposed algorithm for more meaningful models can be considered.

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6 Appendix

6.1 Proof of Proposition 2.1

For the first moment the result follows directly from Equation 2.3 and the martingale property of the Brownian motion and the compensated random measure.

For the second moment we have

$$\begin{aligned} M_2(T) &= \mathbb{E}^T \left[\left(\int_0^t \sigma_{s-} dW_s + \int_0^t \int_{\mathbb{R}} x \tilde{J}(dx, ds) - \frac{1}{2} \int_0^t \sigma_{s-}^2 ds \right)^2 \right] \\ &= \mathbb{E}^T \left[\left(\int_0^t \sigma_{s-} dW_s \right)^2 \right] + \mathbb{E}^T \left[\left(\int_0^t \int_{\mathbb{R}} x \tilde{J}(dx, ds) \right)^2 \right] + \frac{1}{4} \mathbb{E}^T \left[\left(\int_0^t \sigma_{s-}^2 ds \right)^2 \right] \\ &\quad - \mathbb{E}^T \left[\left(\int_0^t \sigma_{s-}^2 ds \right) \left(\int_0^t \int_{\mathbb{R}} x \tilde{J}(dx, ds) \right) \right] - \mathbb{E}^T \left[\left(\int_0^t \sigma_{s-}^2 ds \right) \left(\int_0^t \sigma_{s-} dW_s \right) \right] \\ &\quad + \mathbb{E}^T \left[\left(\int_0^t \int_{\mathbb{R}} x \tilde{J}(dx, ds) \right) \left(\int_0^t \sigma_{s-} dW_s \right) \right] \end{aligned}$$

Because of Itô-isometry for Brownian integrals and for stochastic integrals with respect to random measures, the first equals $\mathbb{E}^T [V_T]$ and the second term equals $\mathbb{E}^T \left[\int_0^t \int_{\mathbb{R}} x^2 J(dx, ds) \right]$. The fourth and fifth term corresponds to the respective covariances due to the martingale properties of the stochastic integrals. The expected value of the product of the two stochastic integrals in the last term is equal to zero because the two underlying processes are assumed to be orthogonal and therefore the integrals with respect to them are also orthogonal. Combining these results completes the proof.

6.2 Proof of Proposition 2.2

The proof can be provided either by using the option-implied risk-neutral densities given in Breeden and Litzenberger (1978) and integration by parts or by the spanning results from Carr and Madan (2001). Here, we make use of the latter. Following Carr and Madan (2001), any twice differentiable payoff $f(S_T)$ of the terminal asset price S_T can be replicated by a static portfolio of European options:

$$f(S_T) = f(K_0) + f'(K_0)(S_T - K_0) + \int_0^{K_0} f''(K)(K - S_T)^+ dK + \int_{K_0}^{\infty} f''(K)(S_T - K)^+ dK$$

for any fixed K_0 . The no-arbitrage price is then given by

$$\begin{aligned} B_0(T)\mathbb{E}^T[f(S_t)] &= f(K_0)B_0(T) + f'(K_0)(S_0 - K_0B_0(T)) \\ &\quad + \int_0^{K_0} f''(K) \text{Put}(S_0, K, T) dK \\ &\quad + \int_{K_0}^{\infty} f''(K) \text{Call}(S_0, K, T) dK \end{aligned}$$

For $f(s) = \ln \frac{sB_0(T)}{S_0} = \ln \frac{s}{F_0}$ and $K_0 = F_0 = \frac{S_0}{B_0(T)}$ we have

$$B_0(T)\mathbb{E}^T \left[\ln \frac{s}{F_0} \right] = - \int_0^{F_0} \frac{\text{Put}(S_0, K, T)}{K^2} dK - \int_{F_0}^{\infty} \frac{\text{Call}(S_0, K, T)}{K^2} dK$$

and for $f(s) = \left(\ln \frac{s}{F_0} \right)^2$

$$\begin{aligned} B_0(T)\mathbb{E}^T \left[\left(\ln \frac{s}{F_0} \right)^2 \right] &= 2 \left(\int_0^{F_0} \frac{(1 - \ln \frac{K}{F_0}) \text{Put}(S_0, K, T)}{K^2} dK \right. \\ &\quad \left. + \int_{F_0}^{\infty} \frac{(1 - \ln \frac{K}{F_0}) \text{Call}(S_0, K, T)}{K^2} dK \right) \end{aligned}$$

Dividing by $B_0(T)$ gives the respective results.